Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Microprocessor & Computer Architecture

BCA — Semester II — 2023 — Answer Sheet

Subject Code: CACS155

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Define addressing mode? Suppose content of register A and B is 37H and 57H respectively. Find the content of flag register after ADD B is performed.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is the requirement of common bus System? Explain with figure.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Explain Symbolic Microinstruction and microinstruction format.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is stack? Give the organization of register stack with all necessary elements.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is pipeline conflict? Explain data dependency and handling of branch instruction in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain the basic working principle of the Control Unit with timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Explain the following instructionsa) LHLD 5050Hb) CPI 34Hc) JNZ 6080H

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Draw the timing diagram for MOV B. For ten bytes of data starting from 7050H, write a program to sort the reading in ascending order.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain the steps of Address Sequencing in detail. Draw and explain selection of address for control memory.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Discuss four-segment instruction pipeline with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
